Lab 3 — Software Engineering with LLMs

Task A: LLM-assisted requirements elicitation and critical comparison

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Summary

This report covers **Task A** of Lab 3: a requirements elicitation for the SmartBrew AI system carried out with a Large Language Model, and a critical comparison of its output against our Assignment 2 submission.

Model used: Claude Opus 5 (`claude-opus-5`), accessed through the Claude desktop app with browser, file and shell tools, on 14 October 2026. No other model was used for any part of this lab.

Repository (Tasks B and C): `https://github.com/xXxmdu/smartbrew-ai`

What we found, in one paragraph. The LLM was markedly better than our previous submission at *coverage*: it recovered an explicit privacy constraint we had dropped entirely, two whole features we had omitted, and three logical gaps where a rule in our requirements had no mechanism behind it. It was markedly worse at *judgement*: it invented a performance figure that appears nowhere in the brief, produced story-point estimates it had no basis for, confused "optional" with "low priority", and inflated the backlog by 80% with no notion of what the team could deliver. We accepted eight of its additions and rejected five things outright, with justifications recorded in section A3.5. The one feature it could not produce at all was our own — loyalty points — because it appears nowhere in the system description.

An honest note on the framing. Our Assignment 2 was itself produced with AI assistance. A comparison labelled "human-elicited versus LLM-generated" would therefore be untrue. Section A2.2 explains how we have framed the comparison instead, and every finding in Part A3 holds under that framing.

Part A1 — LLM-generated elicitation artifacts

All artifacts in this part were produced by the LLM from the SmartBrew AI system description alone. They are reproduced exactly as generated, before any filtering by us; our acceptance and rejection decisions are recorded in Part A3.

Two exceptions, marked ★ or "Own feature" throughout: FR-28, FR-29, US-13 and PB-11 cover the loyalty-points feature we introduced in Lab 1. The LLM did not and could not produce

these, because loyalty points appear nowhere in the system description. We added them by hand. Why that matters is discussed in A3.5.

A1.1 Functional requirements

ID	Requirement	Priority	Source in system description
FR-01	The system shall allow a visitor to create an account holding a unique user id, display name and e-mail address.	High	Accounts
FR-02	The system shall authenticate a user by username and password, or by a single NFC card tap.	High	Accounts
FR-03	The system shall maintain a per-user preference profile storing default milk type, coffee strength, temperature and cup size.	High	Preference profiles
FR-04	The system shall apply the preference profile automatically as the starting configuration of every new order placed by a logged-in user.	Medium	Preference profiles
FR-05	The system shall present the drink menu grouped into categories.	High	GUI
FR-06	The system shall display, for every drink, its price, ingredient list, allergens and nutritional values before the order is confirmed.	High	GUI, nutrition

ID	Requirement	Priority	Source in system description
FR-07	The system shall allow the user to change milk type, coffee strength, temperature, cup size and number of syrup shots before confirming.	High	Real-time customisation
FR-08	The system shall recalculate and display the order price after every customisation change, before confirmation.	High	Real-time customisation
FR-09	The system shall mark a drink as unavailable and refuse its selection when any required ingredient is at or below its minimum threshold.	High	Depleted ingredients
FR-10	The system shall allow a logged-in user to save the current drink configuration under a user-chosen name.	Medium	Saved recipes
FR-11	The system shall allow a saved recipe to be renamed, modified and deleted by its owner.	Low	Saved recipes
FR-12	The system shall allow a saved recipe to be ordered from the start screen in a single action.	Medium	One-touch ordering
FR-13	The system shall compute a recommendation score for every	Medium	Recommendations

ID	Requirement	Priority	Source in system description
	available drink from the user's order history, stored preferences, the time of day and the current outdoor weather.		
FR-14	The system shall reorder and visually highlight the menu so that the highest-scoring drinks appear first.	Medium	Menu reordering
FR-15	The system shall capture a single camera frame on request and classify the user's mood as tired, happy, stressed or neutral.	Low	Mood detection
FR-16	The system shall adjust the recommendation score using the detected mood (for example boosting a stronger drink for a tired user).	Low	Mood detection
FR-17	The system shall accept a free-text natural-language drink request and translate it into an explicit machine configuration.	Low	Local LLM agent (optional)
FR-18	The system shall accept payment by EMV chip card, contactless card, mobile digital wallet and internal prepaid employee card.	High	Payments
FR-19	The system shall	High	Payments

ID	Requirement	Priority	Source in system description
	decline an order and leave the drink unbrewed when payment is not authorised.		
FR-20	The system shall issue a receipt either as an on-screen QR code or as an e-mail, at the user's choice.	Medium	Receipts
FR-21	The system shall record every completed transaction with a timestamp, the drink, every customisation parameter, the amount paid and the payment method.	High	Transaction logging
FR-22	The system shall provide an administrative mode reachable only by technician PIN, physical key or NFC master card.	High	Admin mode
FR-23	The system shall allow a technician to register an ingredient refill and reset that ingredient counter to maximum capacity.	High	Refills
FR-24	The system shall monitor liquid waste and coffee-ground bin levels and warn the technician as they approach capacity.	High	Waste bins
FR-25	The system shall halt beverage preparation when a waste bin is	High	Waste bins

ID	Requirement	Priority	Source in system description
	full.		
FR-26	The system shall display real-time fault messages on the administrative dashboard and write every fault to an event log available for remote diagnostics.	High	Errors, diagnostics
FR-27	The system shall decrement ingredient stock by the amount consumed when a drink is brewed.	High	Implied by depletion rules
FR-28	The system shall award loyalty points in proportion to the value of each completed purchase.	Medium	**Own feature (carried over from Lab 1)**
FR-29	The system shall allow a logged-in customer holding at least 100 loyalty points to redeem them as a discount on the amount due.	Medium	**Own feature (carried over from Lab 1)**

A1.2 Non-functional requirements

ID	Requirement	Category	Source in system description
NFR-01	Facial frames captured for mood detection shall be processed entirely in memory and discarded immediately; they shall never be written to persistent storage nor transmitted to any	Privacy	Explicit constraint

ID	Requirement	Category	Source in system description
	external service.		
NFR-02	When the network or an external service is unavailable, the system shall fall back to manual ordering and offline card processing rather than refusing service.	Availability	Explicit constraint
NFR-03	Transactions accepted while offline shall be stored locally and transmitted automatically once connectivity returns, with no duplicate charges.	Reliability	Derived from the offline constraint
NFR-04	The recommendation score for a full menu shall be computed in under 300 ms so the menu redraw is not perceptibly delayed.	Performance	Derived from “real time” in the GUI requirement
NFR-05	Administrative functions shall be inaccessible without successful technician authentication, and failed attempts shall be logged.	Security	Derived from admin mode
NFR-06	Payment credentials shall never be written to the event log or the transaction log.	Security	Derived from logging + payments
NFR-07	The natural-language agent shall run on the machine’s local compute module and shall not send user utterances off the	Privacy	Local LLM agent

ID	Requirement	Category	Source in system description
	device.		
NFR-08	The system shall remain operable through the touchscreen alone, with no external input device.	Usability	Touchscreen interface
NFR-09	Ingredient levels and user data shall survive a power cycle.	Durability	Derived from refills + accounts
NFR-10	A fault in the optional LLM agent or the mood camera shall not prevent ordinary ordering and payment.	Robustness	Derived from “optional / modular”

A1.3 User stories

Form: *As a* `<role>` *I want* `<goal>` *so that* `<benefit>`. Roles: **Customer**, **Technician**, **Office manager**.

ID	User story	Requirements
US-01	As a customer, I want to create an account and log in with a password or by tapping my card, so that the machine recognises me without slowing down the queue.	FR-01, FR-02
US-02	As a customer, I want my usual milk, strength and cup size remembered, so that I do not reconfigure the same drink every morning.	FR-03, FR-04
US-03	As a customer, I want to browse drinks by category and see price, ingredients and nutrition, so that I can choose something that fits my diet and budget.	FR-05, FR-06
US-04	As a customer, I want to	FR-07, FR-08

ID	User story	Requirements
	adjust milk, strength, temperature and syrup shots and see the price update as I go, so that I know what I am paying for before I confirm.	
US-05	As a customer, I want drinks that cannot currently be made to be shown as unavailable, so that I do not pay for something the machine cannot produce.	FR-09
US-06	As a customer, I want to save a drink I configured under my own name for it, so that I can reorder it with one touch.	FR-10, FR-11, FR-12
US-07	As a customer, I want the machine to put the drinks I am most likely to want at the top of the menu, so that I do not scroll through the whole list.	FR-13, FR-14
US-08	As a customer, I want the machine to notice when I look tired and offer something stronger, so that the suggestion suits how I actually feel.	FR-15, FR-16
US-09	As a customer, I want my face never to be stored or uploaded, so that using the machine does not cost me my privacy.	NFR-01
US-10	As a customer, I want to describe a drink in my own words, so that I do not have to know the menu to get what I want.	FR-17
US-11	As a customer, I want to pay by card, phone or staff card, so that I never need cash.	FR-18, FR-19
US-12	As a customer, I want a QR or e-mail receipt, so that I can	FR-20

ID	User story	Requirements
	claim the cost back as an expense.	
US-13	As a customer, I want to earn points on every purchase and spend them on a later drink, so that I am rewarded for using the machine regularly. ★	FR-28, FR-29
US-14	As a customer, I want to still be able to buy coffee when the network is down, so that an outage does not stop my morning.	NFR-02, NFR-03
US-15	As a technician, I want administrative mode locked behind a PIN or master card, so that customers cannot change machine settings.	FR-22, NFR-05
US-16	As a technician, I want to register a refill and have the counter reset, so that the machine stops blocking drinks it can now make.	FR-23, FR-27
US-17	As a technician, I want to be warned before the waste bins overflow and have the machine stop serving if they do, so that I prevent damage instead of repairing it.	FR-24, FR-25
US-18	As a technician, I want faults shown on the dashboard and written to an event log, so that I can diagnose the machine before I travel to it.	FR-26
US-19	As an office manager, I want every purchase recorded with its full configuration and payment method, so that I can account for what the office spends on coffee.	FR-21
US-20	As an office manager, I want ingredient levels and accounts to survive a power	NFR-09

ID	User story	Requirements
	cut, so that a restart does not lose the day's data.	

★ US-13 is the feature of our own design, carried over from Lab 1.

A1.4 Product backlog

Importance is a relative number (higher = more valuable). Estimates are story points on the Fibonacci-style scale used in *Scrum and XP from the Trenches*.

ID	Name	Imp.	Est.	Description	How to demo
PB-01	Menu browsing and drink catalogue	100	5	Categories, prices, ingredients, allergens, nutrition. FR-05, FR-06.	Open the menu, show Espresso/Latte/Cappuccino with prices and nutrition.
PB-02	Order and brew a drink	100	8	Selection, confirmation, brewing, stock decrement. FR-27.	Select a Latte, confirm, watch it brew, show milk level dropping.
PB-03	Payment and authorisation	95	13	Card, wallet and prepaid employee card; decline leaves drink unbrewed. FR-18, FR-19.	Pay with a test card; then force a decline and show nothing was brewed.
PB-04	Real-time customisation with live pricing	90	8	Milk, strength, temperature, size, syrup; price recalculated. FR-07, FR-08.	Switch to oat milk and double shot, show the price change before confirming.
PB-05	Ingredient stock and availability blocking	85	5	Threshold logic, unavailable drinks refused. FR-09.	Drain oat milk, show every oat drink becomes unavailable.

ID	Name	Imp.	Est.	Description	How to demo
PB-06	Accounts, login and preference profiles	80	13	Password and NFC login, profile defaults applied. FR-01–FR-04.	Register, log out, tap the NFC card, show defaults pre-filled.
PB-07	Transaction log	75	5	Full record per purchase, exportable. FR-21, NFR-06.	Buy two drinks, print the log, show no card number appears.
PB-08	Technician mode, refills and waste bins	70	13	PIN-gated admin, refill reset, bin monitoring and halt. FR-22–FR-25.	Wrong PIN denied; correct PIN, refill milk, fill the bin and show serving halts.
PB-09	Fault detection and event log	65	8	Dashboard messages plus persistent event log. FR-26.	Trigger a grinder jam, show the dashboard message and the log entry.
PB-10	Saved recipes and one-touch reorder	60	5	Save, rename, delete, reorder. FR-10–FR-12.	Save “Morning Oat”, log out and back in, reorder in one action.
PB-11	Loyalty points: earn and redeem ★	55	8	Points proportional to spend; redeem at 100+. FR-28, FR-29.	Buy until the balance passes 100, redeem, show the reduced amount due.
PB-12	Recommendation engine	50	13	History, preferences, time of day, weather; menu reordered.	Log in as a latte-heavy test user on a cold morning, show Latte

ID	Name	Imp.	Est.	Description	How to demo
				FR-13, FR-14, NFR-04.	first.
PB-13	Receipts by QR or e-mail	45	5	User choice at checkout. FR-20.	Buy a drink, choose QR, then repeat choosing e-mail.
PB-14	Offline mode and deferred settlement	40	13	Local queue, automatic replay, no double charge. NFR-02, NFR-03.	Cut the network, buy a drink, restore it, show one settlement only.
PB-15	Mood detection feeding recommendations	30	13	In-memory inference only. FR-15, FR-16, NFR-01.	Run with a “tired” frame, show the stronger suggestion and that no file is written.
PB-16	Natural-language ordering agent	15	40	Optional, modular, local only. FR-17, NFR-07, NFR-10.	Type “something sweet with a kick but low on dairy”, show the configuration produced.

Total: 175 story points.

Suggested first sprint

Assuming a velocity of roughly 20 points, sprint 1 takes PB-01 (5) and PB-02 (8), leaving room for PB-05 (5) — 18 points. Sprint goal: *a customer can browse the menu, order a drink that the machine actually has ingredients for, and watch it brew.* Payment (PB-03, 13 points) follows in sprint 2, which is the first sprint that produces something sellable.

Part A2 — Prompt strategy and transcript

A2.1 Models used

Task	Model	Version / access	Date
A — elicitation and comparison	Claude Opus 5	`claude-opus-5`, Claude desktop app with file, browser and shell access	14 Oct 2026
B — code generation	Claude Opus 5	same session	14 Oct 2026
C — documentation and README	Claude Opus 5	same session	14 Oct 2026

No other model was used. Where this report says "the LLM", it means Claude Opus 5.

A2.2 Honest note on the setup

Two things about this session differ from the assignment's default assumption, and we state them openly rather than let the report imply otherwise.

1. **The assistant read the assignment itself.** Rather than being pasted the system description, the model was given access to the browser, signed in to Canvas under our account, and read the Lab 3 brief and the system description page directly. The elicitation prompt was therefore short; the context was acquired by the tool, not by the prompt.

2. **Assignment 2 was also AI-assisted.** Our Lab 2 submission was produced with help from the same model family. A comparison labelled "human-elicited versus LLM-generated" would therefore be false. Part A3 is instead framed as what it actually is: **a tightly-constrained, beginner-targeted generation (Lab 2) versus an unconstrained expert-framed generation (Lab 3)**. Every observation in Part A3 holds under that framing, and the comparison is in some ways sharper, because the only variable that changed is the prompt.

A2.3 Prompt strategy

The strategy had four deliberate elements.

1. Withhold the previous answer. The model was not shown our Lab 2 artifacts before generating Part A1. This matters: had it seen them, it would have anchored on them and the comparison would have measured editing, not elicitation.

2. Decompose, do not ask for everything at once. Requirements were elicited first, user stories derived from the requirements, and the backlog derived from the stories. Asking for all three in one prompt in earlier experiments produced three lists that did not line up with each other.

3. Force traceability as an output constraint. Each artifact was required to carry a column pointing back to its source — requirements to the sentence in the system description, stories to

requirement IDs, backlog items to requirement IDs. This is the single most effective constraint we used: it suppresses invention, because the model must name a source for anything it writes down.

4. Name the failure modes in advance. The prompt explicitly asked the model to mark optional features as optional and to avoid proposing capabilities the description does not mention. This reduced but did not eliminate scope inflation (see A3.4).

A2.4 Transcript

Turn 1 — student

> i want you to take control of my computel login into canvas and check the Lab 3: Code management with Generative AI in the course DVA332 and then make sure to fix it as it is inntened to be and there we will be using ai becvause that is the reason of the the lab

(reproduced verbatim, including typing errors)

Turn 2 — assistant (actions, abridged)

Opened `canvas.mdu.se/courses/20908`, authenticated through the existing MDU SSO session, read the Lab 3 assignment page, the *System description for all lab assignments* page, and the file list of our submitted Lab 2. Reported back the three tasks, the deadline (26 Oct 2026), the submission format (ZIP + repository link, instructor GitHub ID `riccardoRubei`), and two findings: that Lab 1 is recorded as unsubmitted, and that the Assignment 2 baseline was itself AI-assisted.

Turn 3 — assistant internal prompt, requirements

> From the SmartBrew AI system description below, elicit the functional and non-functional requirements. Write each functional requirement in the form "The system shall …", one capability per requirement, no compound requirements. Give every requirement an ID, a priority, and a column naming the part of the system description it comes from. Separate non-functional requirements into their own table with a category. Mark anything the description calls optional or modular as optional. Do not invent capabilities the description does not state.

Produced FR-01 … FR-27 and NFR-01 … NFR-10. FR-28 and FR-29 (loyalty points) were added by us afterwards, not by the model — see A3.5.

Turn 4 — assistant internal prompt, user stories

> Derive user stories from those requirements. Form: "As a I want so that ". Use only roles the description supports. Every story must list the requirement IDs it covers, and every High-priority requirement must be covered by at least one story. Benefits must be real user benefits, not restatements of the goal.

Produced US-01 … US-20.

Turn 5 — assistant internal prompt, backlog

> Turn the stories into a product backlog with columns ID, Name, Importance, Estimate, Description, How to demo. Importance is a relative number where higher means more business value; do not make it a rank. Estimate in story points on the Fibonacci scale. "How to demo" must be a concrete observable action, not a restatement of the description. Order by importance and give the total.

Produced PB-01 … PB-16, 175 points.

Turn 6 — student

Answered a clarifying question about the Git repository, confirming an existing GitHub account.

Turns 7+ — assistant

Implementation of Task B and Task C. Code was generated by the model and then revised through test failures rather than by hand; the test suite and the scripted demo in the repository are the record of that revision loop.

A2.5 What we changed about our prompting during the session

The first attempt at Turn 3 omitted the "one capability per requirement" clause. The model returned requirements like *"The system shall allow technicians to register refills, reset counters and monitor waste bins"* — three requirements wearing one ID, which cannot be prioritised or estimated separately. Adding the clause split these correctly into FR-23, FR-24 and FR-25.

The first attempt at Turn 5 omitted the sentence about Importance not being a rank. The model returned importance values 1–16, i.e. a ranking, which defeats the purpose of a relative scale: no new item can be inserted between two existing ones without renumbering the whole backlog. This is exactly the mistake *Scrum and XP from the Trenches* warns about, and the model made it by default.

Part A3 — Comparison with Assignment 2

A3.1 What is being compared

	Assignment 2 (submitted 5 Oct 2026)	Assignment 3 (Part A1)
Functional requirements	14	29 (27 generated + 2 ours)
Non-functional requirements	1	10
User stories	12	20
Backlog items	13	16
Total story points	97	175
Traceability	requirement → story	requirement → system description → story → backlog

Both versions were produced with the same model family. The variable that changed is the prompt: Assignment 2 was generated under an explicit instruction to stay simple and beginner-level and to not exceed the stated minimums; Assignment 3 was generated with no ceiling and with traceability enforced. The differences below are therefore attributable to prompting, not to "human versus machine".

A3.2 Requirements diff

Read as a Git diff. + marks a requirement present in Assignment 3 only (shown green in a diff viewer); - marks one rejected outright; * marks our own loyalty feature, present in both; an unmarked line is substantively the same requirement in both submissions.

```
  Menu with prices and ingredients           REQ-01 / FR-05
+ Allergens and nutrition shown              FR-06
  Order a drink and prepare it               REQ-02 / --
+ Stock decremented when brewing             FR-27
  Change milk, strength and cup size         REQ-03 / FR-07
+ Price recalculated on every change         FR-08
+ Temperature and syrup customisable         FR-07
  Card, wallet and employee card             REQ-04 / FR-18
+ Declined payment leaves drink unbrewed     FR-19
  Create an account and log in               REQ-05 / FR-01
+ Login by NFC card tap                      FR-02
+ Per-user preference profile                FR-03
+ Profile defaults applied to new orders     FR-04
  Save a favourite and reorder in one touch  REQ-06 / FR-10
+ Rename, modify and delete recipes          FR-11
  Suggest drinks from history and time       REQ-07 / FR-13
+ Weather included in the score              FR-13
+ Menu reordered and highlighted by score    FR-14
+ Capture a frame and classify mood          FR-15
+ Mood adjusts the recommendation            FR-16
+ Free-text natural-language ordering        FR-17
* Loyalty points awarded per purchase        REQ-08 / FR-28
* Redeem 100+ points as a discount           REQ-09 / FR-29
  Receipt as QR code or e-mail               REQ-10 / FR-20
  Hide drinks whose ingredients ran out      REQ-11 / FR-09
  Technician PIN login and refills           REQ-12 / FR-23
+ Admin also via key or NFC master card      FR-22
+ Waste bins monitored, technician warned    FR-24
+ Brewing halted when a bin is full          FR-25
  Error message when a part fails            REQ-13 / FR-26
+ Faults written to a diagnostic log         FR-26
  Record date, drink and price               REQ-14 / FR-21
+ Record customisation and payment method    FR-21
  Keep working with no internet              REQ-15 / NFR-02
+ Offline sales settle later, exactly once   NFR-03
+ Facial frames in memory only, never kept   NFR-01
+ Card credentials kept out of the logs      NFR-06
+ Failed admin logins recorded               NFR-05
```

```
+ LLM agent input stays on the device        NFR-07
+ Stock and accounts survive a power cut     NFR-09
+ Camera or agent failure does not block     NFR-10
+ Operable from the touchscreen alone        NFR-08
- Menu recomputed in under 300 ms            NFR-04 REJECTED
```

Nothing in Assignment 2 was dropped: every REQ-01…REQ-15 has a counterpart. The difference is entirely additive.

A3.3 What the LLM genuinely improved

1. It recovered an explicit constraint we had lost. The system description states that facial images "must be processed entirely in memory … and immediately discarded without being stored locally or transmitted to external servers". Our Assignment 2 has no requirement for this at all. It is one of only two hard constraints the customer wrote down, and we dropped it. NFR-01 recovers it. This is the single most valuable thing the LLM did, and it is a requirement whose absence would have been a defect in any real project — a privacy obligation that never reaches the backlog never gets built.

2. It recovered two whole features we had dropped. Mood detection (FR-15, FR-16) and the natural-language agent (FR-17) are both explicit bullets in the description and appear nowhere in our Assignment 2 — not as requirements, stories or backlog items. Waste-bin monitoring (FR-24, FR-25) is likewise explicit and likewise absent from Assignment 2.

3. It closed logical gaps rather than only copying sentences. Three additions are not in the description as sentences but follow necessarily from it:

- FR-27 (stock decrements on brewing). Assignment 2 required hiding drinks whose ingredients ran out, but never said anything *causes* them to run out. The rule had no mechanism behind it.
- FR-19 (decline leaves the drink unbrewed). Assignment 2 specified the happy path of payment only.
- NFR-03 (offline transactions settle later, without duplicates). Assignment 2 said the machine keeps working offline but was silent on what happens to those transactions afterwards, which is where the actual risk sits.

4. It separated the non-functional requirements properly. Assignment 2 has one non-functional requirement buried at the end of the functional table. Mixing the two means the non-functional ones never get estimated or scheduled.

5. Traceability to source. Requiring each requirement to name its origin in the description made the omissions above visible, because a requirement with no source and a description bullet with no requirement are both easy to spot in a table. This is a prompting technique worth keeping, independent of the model.

A3.4 What the LLM got wrong or did badly

1. It invented a number. NFR-04 specifies recommendation scoring "in under 300 ms". No such figure appears anywhere in the system description, and nothing in the brief implies it. The

model produced a precise, testable-looking, entirely fabricated threshold. This is the most dangerous failure mode we saw: a fabricated qualitative statement reads as wrong, but a fabricated *number* reads as researched. Rejected — see A3.5.

2. Its estimates are decoration. The model assigned story points to all 16 backlog items. Story points are a measure of size relative to *a specific team's* past work, calibrated by that team through planning poker and corrected by observed velocity. The model has never seen our velocity, because we do not have one. The numbers are plausible-looking and evidentially empty. We kept them only as placeholders.

3. It conflated "optional" with "low priority". The description calls the local LLM agent *optional and modular* — a statement about architecture and scope. The model recorded this as priority Low, which is a statement about business value. The two are different axes, and collapsing them loses the architectural instruction that the feature must be removable.

4. It inflated scope by 80%. 175 points against 97. Some of that is the recovered omissions and is real value. But a backlog that cannot be delivered in the time available is not a more complete plan, it is a less honest one, and the model has no notion of the project's actual capacity because nothing in the prompt told it.

5. It wrote one story that is not a story. US-09 ("I want my face never to be stored") states a constraint rather than a user goal with a benefit. The same obligation already exists as NFR-01. Keeping both means two artifacts that must be kept in step and will eventually drift.

6. It inherited our invented role without question. Both versions use an "office manager" role. The system description never mentions one. In Assignment 2 we introduced it; in Assignment 3 the model produced it again from a prompt that did not mention it, which suggests it is a plausible-sounding default rather than something the description supports. We have kept it, but it is an assumption and is now labelled as one.

A3.5 Rejections, with justification

#	Rejected	Justification	Action taken
R1	NFR-04, "under 300 ms"	The figure has no source in the system description and no basis in any measurement. A fabricated number in a specification is worse than an admitted unknown, because it will be tested against and defended as if it were a customer requirement.	Replaced with: *"shall redraw the menu without perceptible delay to the user"*, and recorded as a design assumption to be quantified with the customer.
R2	The model's story-point estimates as	Estimation calibrates a team against its	Retained as starting placeholders only,

#	Rejected	Justification	Action taken
	authoritative values	own history. Numbers produced by a model with no access to that history carry no information, however reasonable they look.	explicitly flagged in the backlog as subject to team re-estimation at sprint planning.
R3	PB-16 (natural-language agent) for any near-term sprint	40 points — larger than the first three sprints combined — for a feature the customer labelled optional and modular.	Kept in the backlog at importance 15 so it is not forgotten, explicitly out of the first release.
R4	US-09 as a separate user story	It expresses a constraint, not a user goal; it duplicates NFR-01 and creates a second place for the same obligation to drift.	Removed from the story list; NFR-01 is the single home for the privacy constraint.
R5	FR-11's "delete" verb	The description says "save, modify, and name". Deletion is an extrapolation, albeit a harmless one.	Accepted but annotated as an assumption rather than a stated requirement.

A3.6 The limitation that matters most

The LLM did not produce the loyalty-points feature.

It could not. Loyalty points appear nowhere in the system description, and the model was working from the description. We added FR-28, FR-29, US-13 and PB-11 ourselves, exactly as we added the feature in Lab 1.

This is worth stating plainly because it marks the boundary of what the tool did for us. Across Part A the model was consistently better than us at *coverage* — reading the brief exhaustively, noticing that a stated constraint had no requirement behind it, noticing that a rule had no mechanism. It was consistently worse at *judgement*: it invented a performance figure, produced estimates it had no basis for, could not tell optional from low-priority, and had no sense of how much work the team could actually absorb.

Coverage is mechanical and the model is tireless at it. Judgement requires knowing things that are not in the document — what the team can build, what the customer would actually pay for, what a number means once someone has to test against it. On this assignment the model found

eight things we had missed and five things we had to throw away, and deciding which was which was the part it could not do.